C5 Kinetics – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 1: Collision Theory and Maxwell–Boltzmann **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Rate of reaction				Change in concentration per unit time
2	Activation energy (E_a)				Minimum energy for a reaction to occur
3	Two requirements for a successful collision				Energy ≥ E_a and correct orientation
4	Catalyst				Increases rate without being used up
5	How a catalyst increases rate				Alternative route with lower E_a
6	x-axis of a Maxwell–Boltzmann distribution				Energy
7	y-axis of a Maxwell–Boltzmann distribution				Number of molecules
8	Area under the Maxwell–Boltzmann curve				Total number of molecules
9	Does the curve touch the x-axis at high energy?				No
10	Does the curve start at the origin?				Yes

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

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AQA A-level Chemistry • Set 2: Factors and Measuring Rate **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Five factors affecting rate				Temperature, concentration, pressure, surface area, catalyst
2	Peak of a Maxwell–Boltzmann curve				Most probable energy
3	Position of mean energy relative to the peak				To the right
4	Peak at higher temperature				Lower and further right
5	Area under the curve at higher temperature				Unchanged
6	Effect of a catalyst on the curve shape				None
7	Rate from a concentration–time graph				Gradient of tangent
8	Mean rate equation				Change in quantity ÷ time
9	Units of rate				mol dm^{-3} s^{-1}
10	Why a 10 °C rise has a large effect on rate				Many more particles with energy ≥ E_a

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.